

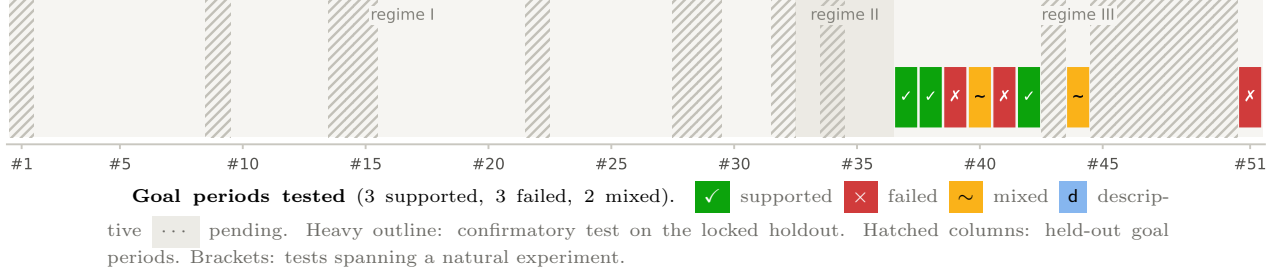
H95: Speed-limit slack as a kickoff-specificity gauge

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 2/5

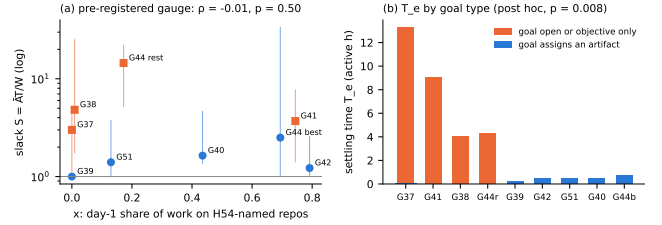
Bottom line. The slack separates assigned-artifact goals from open goals with no overlap, but the HH's specificity measure does not track either; the useful gauge is the goal type, found post hoc and frozen for the holdout.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

I tested whether H75's speed-limit slack $S = \bar{A}T/W$ falls with kickoff specificity across nine regime-III kickoff units (eight periods; G44's two rooms count separately). Specificity x was the HH's: the share of day-1 work on repos that H54 flags as kickoff-named. The synthetic check came first: S falls monotonically with the captured fraction of agents, but nine kickoffs give power 0.29. Result: $\rho(S, x) = -0.01$ ($p = 0.50$). The measure itself fails: a free research week scores $x = 0.74$ and "build your own world" scores 0. Post hoc, coding each goal by whether it assigns a concrete artifact splits the units perfectly: assigned $S = 1.0$ –2.5, open $S = 3.0$ –14.5 ($p = 0.008$). G44's two rooms replicate this on the same days. Holdout: not run.



Observables. (a) Slack against the pre-registered specificity x for the nine kickoff units, with agent-bootstrap 95% CIs (log scale). No trend. Colour is the post-hoc goal type: blue goals assign an artifact, orange goals leave the choice open. (b) Settling time by goal type: assigned goals settle in 0.25–0.75 active h, open goals in 4–13 h.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The operator rule is H75's, now on nine kickoffs: assign each agent (or the group) a concrete artifact in the goal text, and the swarm settles in under an active hour with about one switch per moved agent. Open or objective-only goals take 4–13 active hours and 3–15 switches per net move. The slack is a fine read-out of this after the fact. It is not a thermodynamic cost: the bound is a counting identity. The pre-registered gauge (word overlap between repo names and the goal text) is useless for steering and should not be used.

PHYSICAL MODEL

Models 15 (speed limit) and 11 (kickoff field). Each committing agent is a copy of a jump process on repos plus a null state; $p(t)$ is the share of agents on each repo after the kickoff. W : total-variation distance moved by the settling time T_e . $\bar{A}$: repo switches per agent per active hour. A captured fraction q of agents lands on its settled repo at its first commit; the rest churn between side repos first.

$$T_e \geq W/\bar{A} \quad (\text{counting identity}), \quad S = \bar{A}T_e/W \geq 1$$

$$S \approx \frac{q + (1-q)(1 + a\tau_s)}{q + (1-q)m} \xrightarrow{q \rightarrow 1} 1$$

$$x = \frac{\#\{\text{day-1 commits on named repos}\}}{\#\{\text{day-1 commits}\}}$$

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 $\rho(S, x) \leq -0.5$	$-0.01, p = 0.50$ ($n = 9$; power 0.29)	inconclusive; kill met
P2 $\rho(T_e, x) \leq -0.5$	-0.02	failed
P3 $\rho(S/S_0, x) \leq -0.3$	$+0.60$	failed
P4 naming, not concentration	partial -0.09 ; G39 has $x = 0$	failed
P5 text rival weaker	$\rho(S, S_{\text{text}}) = +0.63$	failed
N1 G44 rooms	$\Delta x = +0.52$; S ratio $\times 5.8$	supported
N2 G38 rooms	$\Delta x = 0.02$: no contrast	failed
Post hoc: goal type	assigned $S \leq 2.5$ (5/5), open $S \geq 3.0$ (4/4); $p = 0.008$	descriptive

GOAL PERIODS IN DETAIL

G37 ✓ supported
G38 ✓ supported
G39 × failed
G40 ~ mixed
G41 × failed
G42 ✓ supported
G44 ~ mixed (units: G44best mixed, G44rest supported)
G51 × failed

FAITHFULNESS SCORECARD

A mapping	0	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

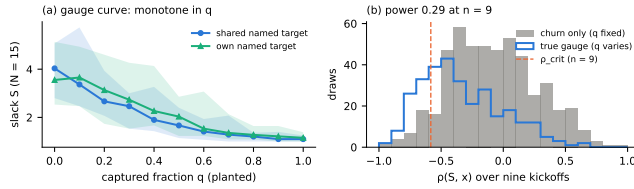
Nine kickoff units cannot give a powered rank test. The goal-type code was chosen after the run, from goal titles I had listed before it; it is frozen for the holdout. G44's #best room has 4 committing agents. G51's slack starts from the null state because its previous week is held out. Commits are the only observation of where agents work, so T_e includes the first-commit delay and has a 15-min floor. S and T_e are not independent of each other.

ROUND 2

Score each agent's own instruction (village plus private goal) for a named artifact and test S against that per-agent field. Merge the slack estimator with H75's in `infra/shared/`. Run `confirm.py` (C2: the four held-out kickoffs are coded open, so each should show $S \geq 2.5$ or $T_e \geq 2$ h).

- **Essence.** a one-number, log-computable gauge of how strongly a goal statement constrains where agents work.
- **H95-R1** A per-agent specificity.
- **H95-R2** More kickoffs.
- **H95-R3** Merge with H75.
- **H95-R4** Run `confirm.py` (C2 is the informative test).

H95-slack-specificity-gauge · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H95-slack-specificity-gauge/README.md · ratings: H95 agent (suggested) (page 2)



Second figure. H75's commit-observed agents with a captured fraction q . (a) The median slack falls monotonically with q for shared and per-agent targets (bands: 10–90%). (b) $\rho(S, x)$ over nine kickoffs: with churn rates varying 16× and q fixed (grey) the test has size 0.06; with q varying (blue) it reaches ρ_{crit} in only 29% of draws.